Appendix: R Code for Crime Analysis

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Appendix: R Code

A.1 Data Loading and Preparation

```
library(tidyverse)
library(lubridate)
library(readr)
library(dplyr)
library(ggplot2)
library(broom)
library(fixest)
library(EValue)

# Load dataset
crime <- read_csv("crime_dataset.csv",
  col_types = cols(DRUGS = col_number())
)

# Parse dates
crime$Date <- dmy(crime$Date)

# Restrict to study period: July 2013 - March 2020
crime <- crime %>%
  filter(Date >= as.Date("2013-07-01") & Date <= as.Date("2020-03-08"))

# Post-intervention indicator (July 1, 2016 onwards)
crime$post <- ifelse(crime$Date >= as.Date("2016-07-01"), 1, 0)
```

A.2 Variable Construction

Treatment Variable

```r
# Drug crime rate (share of total crimes)
crime <- crime %>%
  mutate(
    drug_rate = DRUGS / `GRAND TOTAL`
  )
```

Control Variable

```r
# Aggregate non-drug crimes
crime <- crime %>%
  mutate(
    non_drug_total =
      `THEFT  - RPC Art. 308` +
      `ROBBERY  - RPC Art. 293` +
      `QUALIFIED THEFT  - RPC Art. 310  as amended by BP Blg 71` +
      `HIGHWAY ROBBERY` +
      `SWINDLING (ESTAFA)  - RPC Art. 315 as amended by PD 1689` +
      `OTHER FORMS OF SWINDLING  - RPC Art. 316 as amended by PD 1689` +
      `BOUNCING CHECK` +
      `NEW ANTI-CARNAPPING ACT OF 2016 - MC - RA 10883 (repealed RA 6539)`,
    non_drug_rate = non_drug_total / `GRAND TOTAL`
  )
```

A.3 Difference-in-Differences (DiD) Estimation

Dataset Preparation

```r
# Reshape to long format for DiD
did_data_p <- crime %>%
  select(Date, drug_rate, non_drug_rate, post) %>%
  pivot_longer(
```

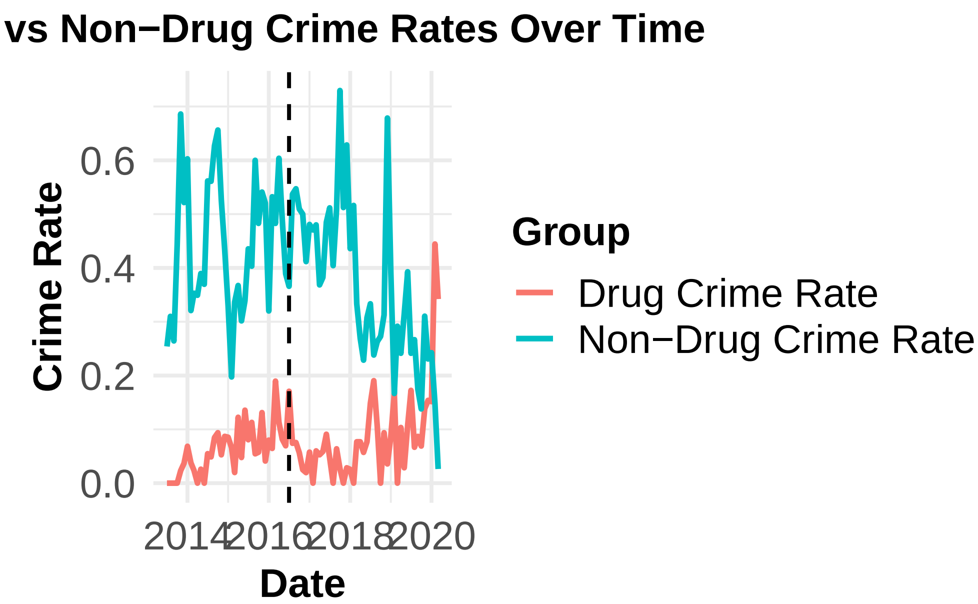
```r
    cols      = c(drug_rate, non_drug_rate),
    names_to  = "group",
    values_to = "crime_rate"
  ) %>%
  mutate(treat = ifelse(group == "drug_rate", 1, 0))
```

DiD Model

```r
did_model_p <- lm(crime_rate ~ treat + post + treat * post,
                  data = did_data_p)
summary(did_model_p)
```

```

Call:
lm(formula = crime_rate ~ treat + post + treat * post, data = did_data_p)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.34259 -0.06084 -0.01001  0.06151  0.36083 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.44177    0.01845  23.939  < 2e-16 ***
treat       -0.38093    0.02610 -14.596  < 2e-16 ***
post        -0.07287    0.02476  -2.943  0.00374 ** 
treat:post   0.09896    0.03501   2.826  0.00532 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1107 on 158 degrees of freedom
Multiple R-squared:  0.6954,	Adjusted R-squared:  0.6896 
F-statistic: 120.2 on 3 and 158 DF,  p-value: < 2.2e-16
```

DiD Plot



A.4 Parallel Trends Assumption Tests

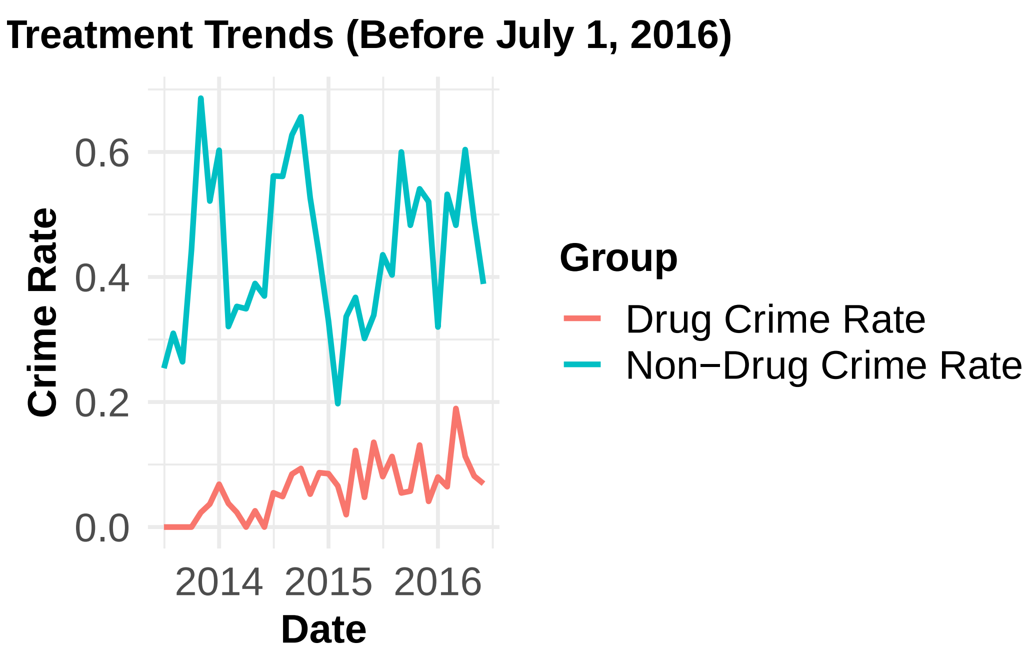
A.4.1 Pre-Trend Regression Test

```
# Restrict to pre-treatment period
pre_data_p <- did_data_p %>%
  filter(Date < as.Date("2016-07-01")) %>%
  mutate(time_num = as.numeric(Date))

# Test for differential pre-trends
pretrend_model_p <- lm(crime_rate ~ treat * time_num, data = pre_data_p)
summary(pretrend_model_p)
```

```
Call:
lm(formula = crime_rate ~ treat * time_num, data = pre_data_p)
```

```
Residuals:
      Min        1Q    Median        3Q       Max 
-0.247254 -0.043590 -0.005005  0.040030  0.269188 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)    -5.558e-01  7.968e-01  -0.698    0.488
treat          -9.389e-01  1.127e+00  -0.833    0.408
time_num        6.075e-05  4.852e-05   1.252    0.215
treat:time_num  3.398e-05  6.861e-05   0.495    0.622

Residual standard error: 0.09202 on 68 degrees of freedom
Multiple R-squared:  0.8219,    Adjusted R-squared:  0.8141 
F-statistic: 104.6 on 3 and 68 DF,  p-value: < 2.2e-16
```

Pre-Trend Plot



A.4.2 Placebo Test

```r
# Use pre-treatment data with a fake intervention date (July 2015)
placebo_data <- did_data_p %>%
  filter(Date < as.Date("2016-07-01")) %>%
  mutate(
    FakeTreat1 = as.integer(group == "drug_rate" & Date >= as.Date("2015-07-01")),
    FakeTreat2 = as.integer(group == "drug_rate" & Date == as.Date("2015-07-01"))
  )

# Estimate placebo effects with two-way fixed effects
pl_model1 <- feols(crime_rate ~ FakeTreat1 | group + Date, data = placebo_data)
pl_model2 <- feols(crime_rate ~ FakeTreat2 | group + Date, data = placebo_data)

etable(pl_model1, pl_model2)
```

```
                        pl_model1       pl_model2
Dependent Var.:        crime_rate      crime_rate

FakeTreat1      -0.0191 (0.0436)
FakeTreat2                        0.0268 (0.1253)
Fixed-Effects:  ---------------- ---------------
group                         Yes             Yes
Date                          Yes             Yes
_______________ ________________ _______________
S.E. type                     IID             IID
Observations                   72              72
R2                        0.92015         0.91980
Within R2                 0.00564         0.00135
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

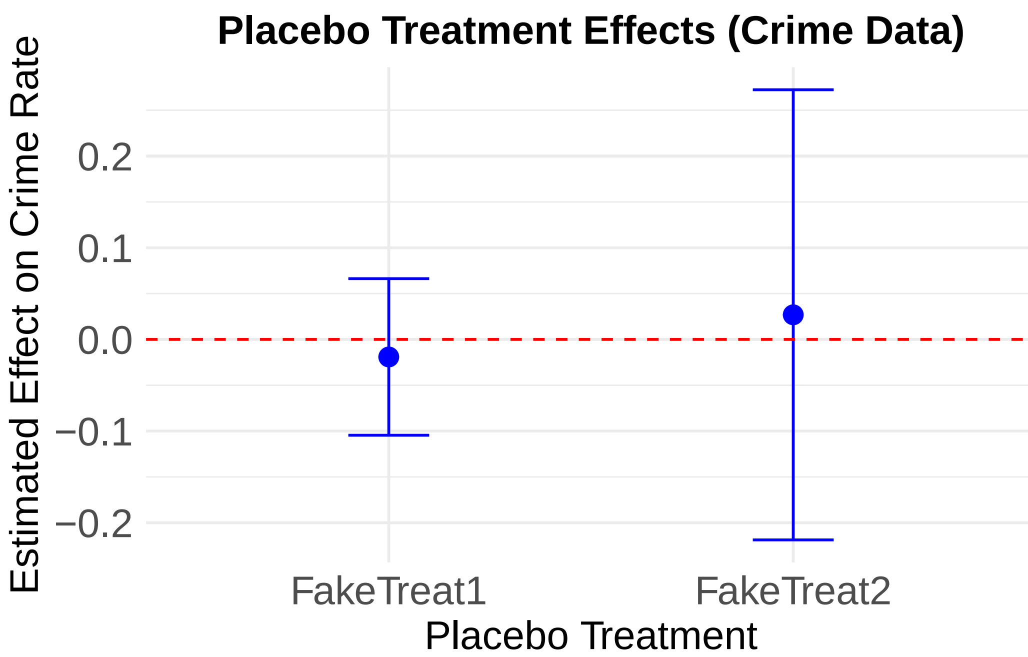
```r
# Collect coefficients and confidence intervals for plot
coef_df <- tibble(
  Model    = c("FakeTreat1", "FakeTreat2"),
  Estimate = c(coef(pl_model1)["FakeTreat1"],
               coef(pl_model2)["FakeTreat2"]),
  SE       = c(summary(pl_model1)$coeftable["FakeTreat1", "Std. Error"],
               summary(pl_model2)$coeftable["FakeTreat2", "Std. Error"])
) %>%
  mutate(Lower = Estimate - 1.96 * SE,
```

```r
         Upper = Estimate + 1.96 * SE)

print(coef_df)
```

```
# A tibble: 2 x 5
  Model       Estimate     SE  Lower  Upper
  <chr>          <dbl>  <dbl>  <dbl>  <dbl>
1 FakeTreat1   -0.0191 0.0436 -0.105 0.0663
2 FakeTreat2    0.0268 0.125  -0.219 0.272
```

Placebo Plot

```r
placebo <- ggplot(coef_df, aes(x = Model, y = Estimate)) +
  geom_point(size = 3, color = "blue") +
  geom_errorbar(aes(ymin = Lower, ymax = Upper), width = 0.2, color = "blue") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Placebo Treatment Effects (Crime Data)",
       y = "Estimated Effect on Crime Rate",
       x = "Placebo Treatment") +
  theme_minimal() +
  theme(
    plot.title   = element_text(size = 15, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 15),
    axis.title.y = element_text(size = 15),
    axis.text.x  = element_text(size = 15),
    axis.text.y  = element_text(size = 15)
  )

ggsave("placebo_plot.png", plot = placebo, width = 8, height = 5, dpi = 300)
placebo
```



A.5 Event Study

```
# Construct July-to-July event-time bins relative to July 2016
did_data_p_ju <- did_data_p %>%
  mutate(
    year_shift = ifelse(month(Date) >= 7, year(Date), year(Date) - 1),
    event_time = factor(year_shift - 2016)
  )

# Set baseline to year -1 (July 2015 - June 2016)
did_data_p_ju$event_time <- relevel(did_data_p_ju$event_time, ref = "-1")

# Event study regression
event_model_ju <- lm(crime_rate ~ treat * event_time, data = did_data_p_ju)
summary(event_model_ju)
```

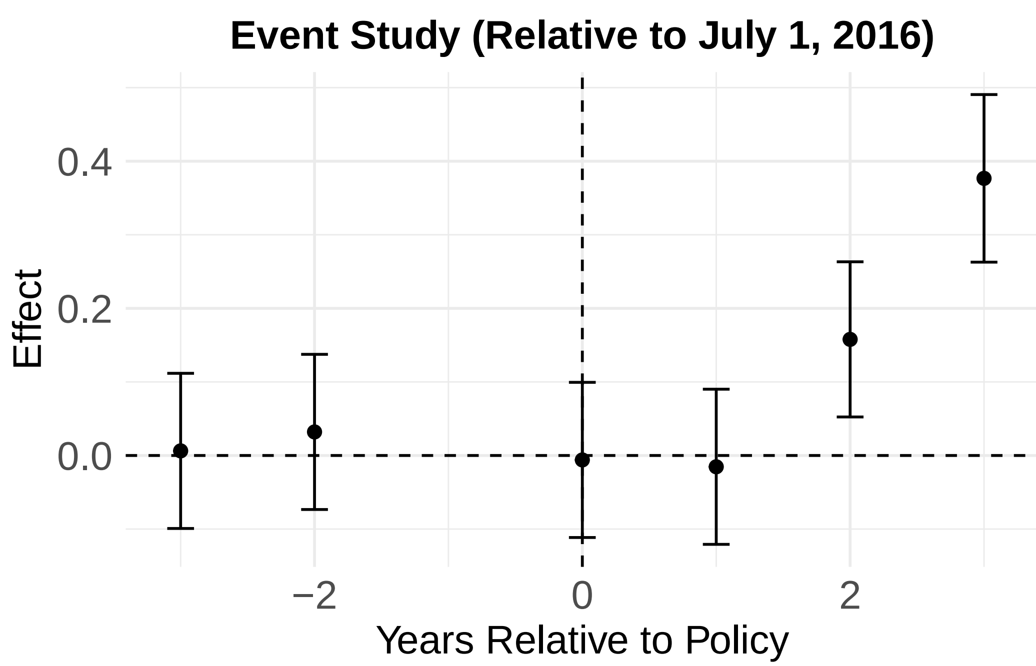
```
Call:
```

```
lm(formula = crime_rate ~ treat * event_time, data = did_data_p_ju)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.23915 -0.04842 -0.00662  0.03815  0.35386 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)         0.483373   0.026901  17.969  < 2e-16 ***
treat              -0.393684   0.038044 -10.348  < 2e-16 ***
event_time-3       -0.077948   0.038044  -2.049  0.04224 *  
event_time-2       -0.046850   0.038044  -1.231  0.22010    
event_time0        -0.021853   0.038044  -0.574  0.56656    
event_time1        -0.034384   0.038044  -0.904  0.36758    
event_time2        -0.158661   0.038044  -4.170 5.16e-05 ***
event_time3        -0.285829   0.041092  -6.956 1.05e-10 ***
treat:event_time-3  0.006243   0.053802   0.116  0.90778    
treat:event_time-2  0.032018   0.053802   0.595  0.55268    
treat:event_time0  -0.006040   0.053802  -0.112  0.91076    
treat:event_time1  -0.015347   0.053802  -0.285  0.77585    
treat:event_time2   0.157817   0.053802   2.933  0.00389 ** 
treat:event_time3   0.376679   0.058113   6.482 1.27e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09319 on 148 degrees of freedom
Multiple R-squared:  0.7979,	Adjusted R-squared:  0.7802 
F-statistic: 44.95 on 13 and 148 DF,  p-value: < 2.2e-16
```

Event Study Plot

```r
event_plot_ju <- tidy(event_model_ju) %>%
  filter(grepl("treat:event_time", term)) %>%
  mutate(event_time = as.numeric(gsub("treat:event_time", "", term)))

event_study <- ggplot(event_plot_ju, aes(x = event_time, y = estimate)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = estimate - 1.96 * std.error,
                    ymax = estimate + 1.96 * std.error), width = 0.2) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
```

```r
  labs(title = "Event Study (Relative to July 1, 2016)",
       x = "Years Relative to Policy",
       y = "Effect") +
  theme_minimal() +
  theme(
    plot.title   = element_text(size = 15, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 15),
    axis.title.y = element_text(size = 15),
    axis.text.x  = element_text(size = 15),
    axis.text.y  = element_text(size = 15)
  )

ggsave("event_study_plot.png", plot = event_study, width = 8, height = 5, dpi = 300)
event_study
```



==

E-VALUE SENSITIVITY ANALYSIS FOR DiD

==

E-Value Input Parameters

```r
est  <- coef(did_model_p)["treat:post"]
print(est)
```

```
treat:post
0.09896335
```

```r
se <- coef(summary(did_model_p))["treat:post", "Std. Error"]
print(se)
```

```
[1] 0.0350149
```

```r
sd_y <- sigma(did_model_p)
print(sd_y)
```

```
[1] 0.1107269
```

E-Value Results

```r
EValue::evalues.OLS(est = est, se = se, sd = sd_y, delta = 1)
```

```
            point    lower    upper
RR       2.255389 1.284587 3.959857
E-values 3.938063 1.889216       NA
```

E-Value Contour Plot

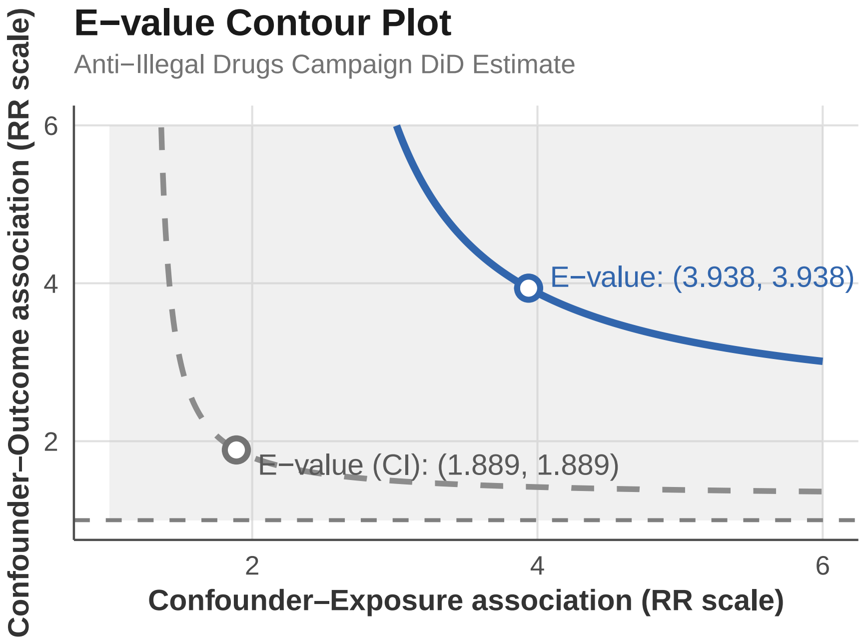
```r
#   1. Crime DiD model
did_model_p <- lm(
  crime_rate ~ treat + post + treat * post,
  data = did_data_p
)

#   2. Extract DiD coefficient, SE, and residual SD
est  <- coef(did_model_p)["treat:post"]
se <- coef(summary(did_model_p))["treat:post", "Std. Error"]
sd_y <- sigma(did_model_p)

#   3. Compute E-values
ev_crime <- EValue::evalues.OLS(
  est   = est,
  se    = se,
  sd    = sd_y,
  delta = 1
)

#   4. Extract RR and E-values
rr_point <- ev_crime["RR", "point"]
rr_lower <- ev_crime["RR", "lower"]

e_point <- ev_crime["E-values", "point"]
e_ci    <- ev_crime["E-values", "lower"]

#   5. E-value contour function
evalue_curve <- function(x, rr) {
  rr * (x - 1) / (x - rr)
}

#   6. Curve data
# Point-estimate contour
df_point <- data.frame(
  x = seq(rr_point + 0.0001, 6.0, length.out = 2000)
)
df_point$y <- evalue_curve(df_point$x, rr_point)
df_point <- subset(df_point, y >= 1 & y <= 6)

# CI contour
df_ci <- data.frame(
```

```r
  x = seq(rr_lower + 0.0001, 6.0, length.out = 2000)
)
df_ci$y <- evalue_curve(df_ci$x, rr_lower)
df_ci <- subset(df_ci, y >= 1 & y <= 6)

#   7. Labels
point_label <- data.frame(
  x = e_point,
  y = e_point,
  label = sprintf("E-value: (%.3f, %.3f)", e_point, e_point)
)

ci_label <- data.frame(
  x = e_ci,
  y = e_ci,
  label = sprintf("E-value (CI): (%.3f, %.3f)", e_ci, e_ci)
)

#   8. Plot
p_crime <- ggplot() +

  # background
  annotate(
    "rect",
    xmin = 1, xmax = 6,
    ymin = 1, ymax = 6,
    fill = "grey80",
    alpha = 0.30
  ) +

  # null line
  geom_hline(
    yintercept = 1,
    linetype = "dashed",
    linewidth = 0.7,
    color = "grey50"
  ) +

  # main E-value contour
  geom_line(
    data = df_point,
    aes(x = x, y = y),
```

```r
  color = "#3266AD",
  linewidth = 1.3
) +

# CI contour
geom_line(
  data = df_ci,
  aes(x = x, y = y),
  color = "grey55",
  linewidth = 1.0,
  linetype = "dashed"
) +

# point estimate marker
geom_point(
  data = point_label,
  aes(x = x, y = y),
  shape = 21,
  size = 3,
  fill = "white",
  color = "#3266AD",
  stroke = 1.6
) +

# point estimate text
geom_text(
  data = point_label,
  aes(x = x, y = y, label = label),
  nudge_x = 0.15,
  nudge_y = 0.15,
  hjust = 0,
  size = 3.8,
  color = "#3266AD"
) +

# CI marker
geom_point(
  data = ci_label,
  aes(x = x, y = y),
  shape = 21,
  size = 3,
  fill = "white",
```

```
  color = "grey45",
  stroke = 1.6
) +

# CI text
geom_text(
  data = ci_label,
  aes(x = x, y = y, label = label),
  nudge_x = 0.15,
  nudge_y = -0.18,
  hjust = 0,
  size = 3.8,
  color = "grey35"
) +

labs(
  title = "E-value Contour Plot",
  subtitle = "Anti-Illegal Drugs Campaign DiD Estimate",
  x = "Confounder-Exposure association (RR scale)",
  y = "Confounder-Outcome association (RR scale)"
) +

coord_cartesian(
  xlim = c(1, 6),
  ylim = c(1, 6),
  clip = "off"
) +

theme_minimal(base_size = 13) +
theme(
  plot.title = element_text(
    face = "bold",
    size = 14,
    hjust = 0,
    color = "grey10",
    margin = margin(b = 4)
  ),
  plot.subtitle = element_text(
    size = 10,
    hjust = 0,
    color = "grey45",
    margin = margin(b = 10)
```

```
    ),
    axis.title = element_text(
      face = "bold",
      size = 11,
      color = "grey20"
    ),
    axis.text = element_text(
      color = "grey30",
      size = 10
    ),
    axis.line = element_line(
      color = "grey30",
      linewidth = 0.4
    ),
    panel.grid.major = element_line(
      color = "grey88",
      linewidth = 0.35
    ),
    panel.grid.minor = element_blank(),
    plot.margin = margin(12, 65, 12, 12)
  )

p_crime
```




```
# Save
ggsave(
  "evalue_did_crime_plot.png",
  plot = p_crime,
  width = 8,
  height = 5,
  dpi = 300,
  bg = "white"
)
```